Corporate credit risk model review

6,398
usable synthetic borrower records

12 months
forward default outcome

Sample	Snapshot period	Borrowers
Training	2017 to 2019	2,400
Maturity embargo	2020 and 2022	1,600
Validation	2021	800
Test	2023 to 2024	1,598

Embargo years allow labels to mature before model selection. All test outcomes mature by the end of 2025.

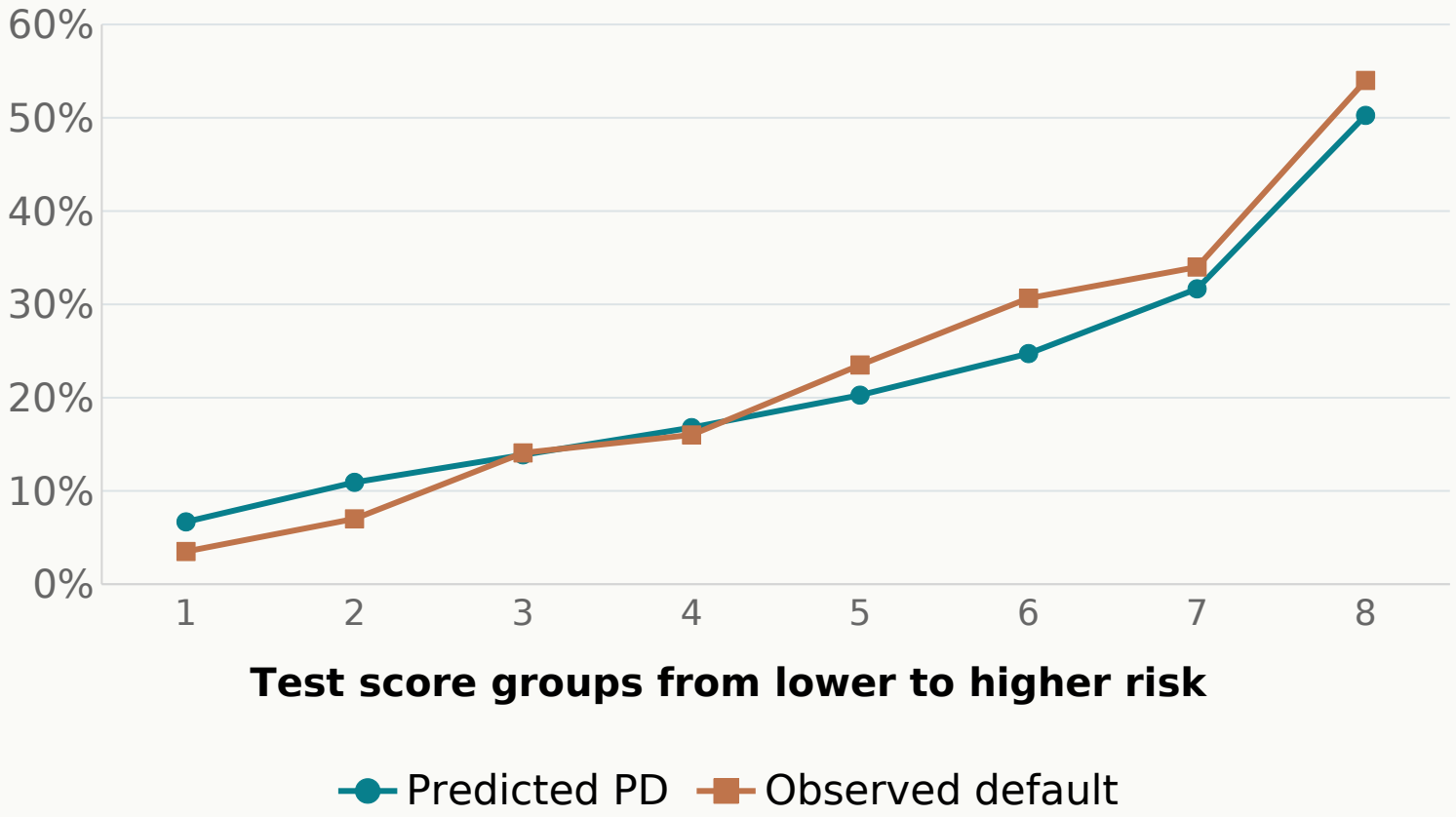
Held out model performance

Metric	Logistic regression	Decision tree
ROC-AUC	0.745	0.711
Average precision	0.454	0.385
Brier score	0.154	0.159
Recall at locked threshold	56.4%	63.0%

Test sample: 1,598 borrowers, 22.8% observed default rate.

Validation F1 selects thresholds of 0.25 for logistic regression and 0.21 for the tree.

Probability calibration by risk group



21.9%

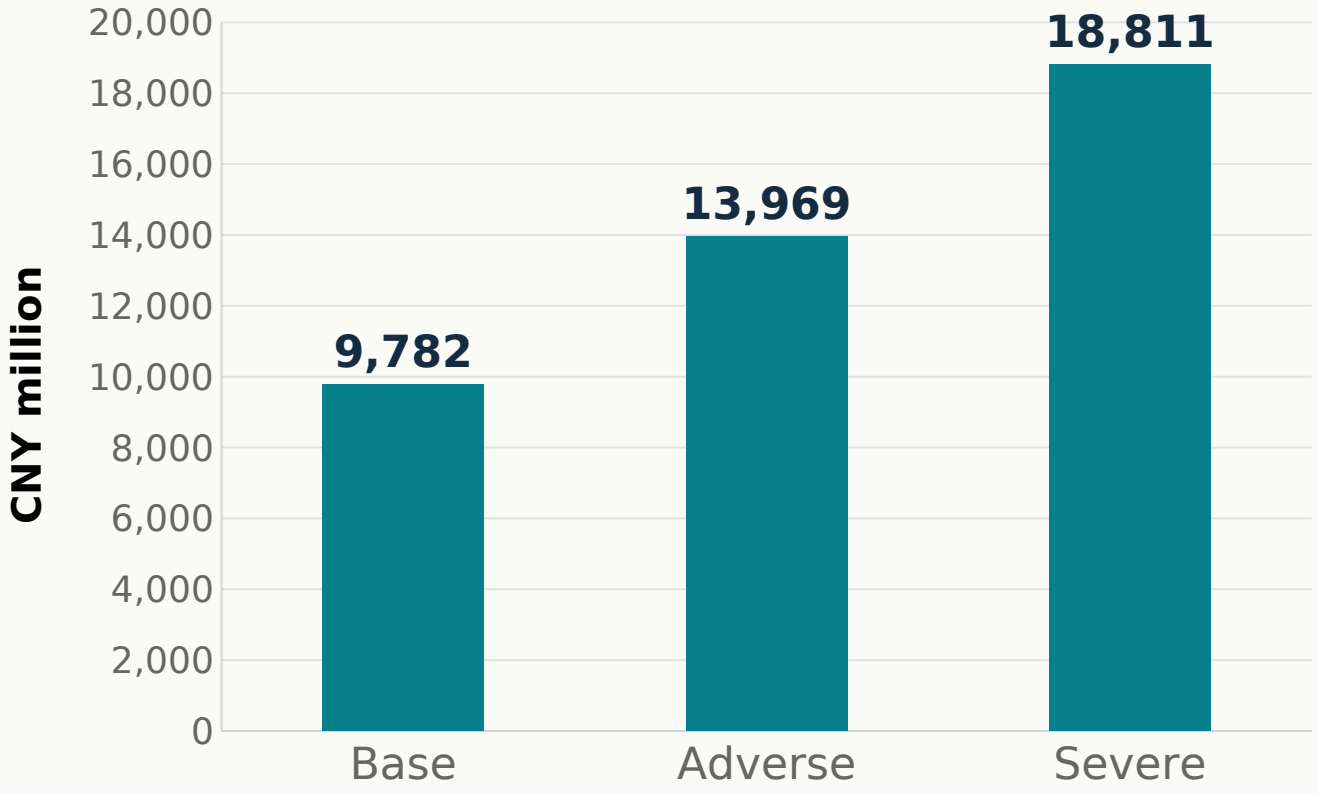
mean predicted PD

22.8%

observed default rate

Calibration matters when probabilities feed expected loss calculations.

Conditional stress loss proxy



Severe assumptions

Leverage rises by 0.10
DSCR falls by 30%
Borrowing rate rises by 2 pp
GDP growth falls by 4 pp

12 month loss = debt × PD × 45% LGD
Illustrative shocks on the fixed test cohort

Mean PD: 21.9% base, 44.2% severe.

Review conclusions and limitations

Model evidence

Temporal separation prevents using future outcomes in training. Validation selects operating thresholds before test evaluation.

HC1 standard errors quantify coefficient uncertainty. Separate VIF diagnostics assess collinearity.

Limits of interpretation

Synthetic generation constrains what these results establish. Shared macro conditions can affect the independence assumed by HC1 inference.

GDP growth is outside the training range for 800 of 1,598 base borrowers and all stressed borrowers. Scenario results rely on extrapolation.